

PIUM EMPIRICAL GALAXY ROTATION CURVE RE-ANALYSIS

Full SPARC Recalculation from Raw Galaxy Rotation Data

Author: Szabolcs Krehlik
ORCID: 0009-0003-8623-7876
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Research Context: Phase–Interference Universe Model (PIUM)
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ABSTRACT

This work presents a full recalculation of galaxy outer rotation-curve behaviour using the SPARC galaxy database. The aim of the study is to establish a transparent and reproducible empirical baseline for outer galaxy dynamics.

The primary analysed quantity is the logarithmic outer slope

$$d \log_{10}(V) / d \log_{10}(R)$$

calculated in the outer regions of galaxies using a scale-length based radial definition.

The analysis includes galaxies with quality flags $Q = 1$ or $Q = 2$ and with at least eight valid outer-region data points.

The resulting analysis sample contains 79 galaxies from the original SPARC catalog of 175 systems.

The median outer slope is -0.000632 , indicating that the typical galaxy in the accepted sample exhibits nearly flat outer rotation behaviour.

This dataset provides a reproducible observational reference that can later be interpreted within the theoretical framework of the Phase–Interference Universe Model (PIUM).

1. INTRODUCTION

Galaxy rotation curves remain one of the most important observational probes of galactic dynamics.

The outer regions of these curves are particularly significant because they probe the large-scale dynamical structure of galaxies.

Before theoretical interpretation can be attempted, the empirical structure of the observations must be reconstructed using a transparent and reproducible method.

The present work performs a full recalculation of outer rotation behaviour using the SPARC galaxy rotation dataset.

Although the broader research program includes the Phase–Interference Universe Model (PIUM), this document is strictly empirical and methodological.

Its objectives are:

- reconstruction of outer rotation behaviour from raw data
- documentation of the full calculation pipeline
- explicit recording of galaxy inclusion and exclusion
- production of a reproducible statistical baseline

2. DATA SOURCES

Primary dataset: SPARC galaxy rotation database.

The SPARC catalog contains rotation curves and photometric measurements for 175 disk galaxies.

The catalog parameters include:

Galaxy name
Distance
Inclination
3.6 μm luminosity
Disk scale length R_{disk}
HI mass
HI radius
Asymptotic rotation velocity V_{flat}
Quality flag Q

Quality flag definition

$Q = 1$ high quality
 $Q = 2$ medium quality
 $Q = 3$ low quality

Primary data repositories used:

SPARC galaxy database
<https://astroweb.case.edu/SPARC/>

SPARC dataset mirror
<https://zenodo.org/records/16284118>

SPARC reference publication
Lelli, F., McGaugh, S. S., & Schombert, J. M. (2016).
SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves.

<https://arxiv.org/abs/1606.09251>

3. SAMPLE SELECTION

Initial SPARC catalog size

175 galaxies

Quality filter

Accepted galaxies: $Q = 1$ or $Q = 2$
Excluded galaxies: $Q = 3$

Outer region definition

$R > 2.5 \times R_{\text{disk}}$

Minimum outer-region requirement

$N_{\text{outer}} \geq 8$

Final sample counts

Accepted galaxies: 79
Excluded galaxies: 96

4. OUTER SLOPE CALCULATION

Outer slope definition

$$s = d \log_{10}(V_{\text{obs}}) / d \log_{10}(R)$$

Regression method

Ordinary Least Squares regression in $\log_{10}$ - $\log_{10}$ space.

Operational classification

Rising galaxy

$s > 0.05$

Flat galaxy

$-0.05 \leq s \leq 0.05$

Declining galaxy

$s < -0.05$

5. MAIN RESULTS

Accepted galaxies: 79

Mean slope
 0.044600

Median slope
 -0.000632

Standard deviation
 0.176476

Minimum
 -0.217107

Maximum
 0.613538

Galaxy classification

Rising galaxies: 33
Flat galaxies: 18
Declining galaxies: 28

6. CORRELATION WITH VFLAT

Outer slope shows a statistically significant negative correlation with V_{flat} .

Pearson correlation coefficient

$r \approx -0.63$

Spearman rank correlation

$\rho \approx -0.66$

This indicates that galaxies with larger asymptotic rotation velocities tend to exhibit flatter or declining outer slopes.

7. RADIAL ACCELERATION RELATION

The radial acceleration relation (RAR) was recalculated using the rotation curve data of the SPARC galaxies.

Two different point counts are relevant.

Total SPARC RAR points

3271

This number corresponds to all available radial data points across the SPARC galaxy sample.

Accepted-sample RAR points

2338

This subset includes only the galaxies that passed the quality and outer-region filters used in the main analysis ($Q \leq 2$ and $N_{\text{outer}} \geq 8$).

RAR regression results for the accepted sample

Slope ≈ 0.66
Correlation coefficient ≈ 0.95
Scatter ≈ 0.169 dex

This confirms that the recalculated dataset reproduces the well-known radial acceleration relation within the accepted galaxy sample.

The regression was performed in $\log_{10}(\text{gobs}) - \log_{10}(\text{gbar})$ space using the accepted galaxy sample.

8. UNCERTAINTY AND ERROR PROPAGATION

Outer slope uncertainty arises from several sources.

Velocity measurement uncertainty

If σ_V represents the velocity error, the logarithmic error becomes

$$\sigma_{\log V} \approx \sigma_V / (V \ln 10)$$

Radius uncertainty

$$\sigma_{\log R} \approx \sigma_R / (R \ln 10)$$

Regression uncertainty

The slope uncertainty is estimated using the standard error of the regression coefficient.

Systematic uncertainty sources

distance uncertainties
inclination corrections
baryonic modelling assumptions
outer-region definition
sample selection thresholds

These uncertainties are documented but not fully propagated galaxy-by-galaxy in the current version.

9. GRAPHICAL ANALYSIS

The following figures accompany the Zenodo dataset.

Figure 1
Outer slope histogram

Figure 2
Outer slope vs V_{flat}

Figure 3
Outer slope vs R_{disk}

Figure 4
Radial Acceleration Relation plot

Figure 5
RAR residual distribution

Figure 6
Outer slope sensitivity comparison

Figure 7
Example rotation curves

All figures are generated from the recalculated dataset.

10. REPRODUCIBILITY

The Zenodo archive accompanying this study contains the following data files.

Main galaxy catalog

pium_sparc_recalc_main_table.csv

Accepted galaxy sample

pium_sparc_accepted_galaxies.csv

Excluded galaxy list

pium_sparc_excluded_galaxies.csv

RAR dataset

pium_sparc_rar_points_q12.csv

Method sensitivity tables

pium_sparc_outer_slope_sensitivity.csv
pium_sparc_method_sensitivity_summary.csv

Computation summary

pium_sparc_recalc_summary.txt

11. RELATION TO PIUM

The Phase–Interference Universe Model proposes that large-scale physical structures may arise from organised field behaviour.

Galaxy rotation curves represent a natural observational domain in which such field-based interpretations could be explored.

The present study does not claim confirmation of PIUM.

Instead it provides a transparent empirical dataset against which PIUM-based interpretations may later be tested.

APPENDIX A FULL GALAXY CATALOG

The complete SPARC galaxy catalog used in this analysis is provided in

pium_sparc_recalc_main_table.csv

Columns include

Galaxy
 Q
Distance
Inclination
 R_{disk}
 V_{flat}
MHI
analysis flags

APPENDIX B ACCEPTED GALAXIES

The final analysis sample (79 galaxies) is listed in

pium_sparc_accepted_galaxies.csv

Columns

Galaxy
 Q
 R_{disk}
 V_{flat}
 N_{outer}
outer_slope
outer_slope_error
flatness_index
stability_index

APPENDIX C EXCLUDED GALAXIES

The list of excluded galaxies and the corresponding exclusion reasons are provided in

pium_sparc_excluded_galaxies.csv

Columns

Galaxy
 Q
 R_{disk}
 N_{outer}
exclusion_reason

APPENDIX D RAR DATASET

The radial acceleration relation dataset used in the analysis is available in

pium_sparc_rar_points_q12.csv

Columns

Galaxy
 R
gobs
gbar

APPENDIX E METHOD SENSITIVITY TABLES

Outer slope sensitivity calculations are provided in

pium_sparc_outer_slope_sensitivity.csv

Summary statistics for the sensitivity tests are provided in

pium_sparc_method_sensitivity_summary.csv

DATA SOURCE REFERENCES

Lelli, F., McGaugh, S. S., & Schombert, J. M. (2016).
SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves.

The Astronomical Journal, 152, 157.
<https://doi.org/10.3847/0004-6256/152/6/157>

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